

HHI_PCB_APD-Array-EVB_V0-1-0 Bring-up Protocol

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1. Objective

To validate the functionality of the **HHI APD-Array Evaluation Board** by verifying the power rails, load-switch sequencing, negative high-voltage generation, logic-level shifting, SPI interface, and ASIC connectivity via the DDR3 connector.

Ensure safe operation of the COMPASS photodiode array ASIC and confirm correct PCB assembly before applying full system power.

2. Test Setup

Input Power Sources

- Bench Power Supply: +12 V, 0..1 A
- USB-C cable (for ESP32-S3 Feather only)

Equipment

- Digital multimeter (DMM)
- Oscilloscope (≥ 100 MHz)
- Current-limited bench supply
- Fine oscilloscope probes for DDR3 connector probing
- VCSEL/laser for illumination testing
- Differential SMA oscilloscope probes

3. Safety & Pre-Checks

- ☒ ~~Visual inspection for solder bridges and IC orientation~~
- ☒ ~~Confirm polarity of diodes (D100, D101, D102) and high voltage capacitors (C113-C121)~~
- ☒ ~~Ensure all 0- Ω links reflect the desired configuration (SPI bus selection, level shifter direction jumpers)~~
- ☒ ~~Check for shorts:~~
 - ☒ ~~+12 V-GND ($> 1\text{ k}\Omega$)~~
 - ☒ ~~+5 V-GND ($> 1\text{ k}\Omega$)~~
 - ☒ ~~3.3 V rails-GND ($> 1\text{ k}\Omega$)~~
 - ☒ ~~1.8 V rails-GND ($> 1\text{ k}\Omega$)~~
 - ☒ ~~-60 V node floating (no short)~~

4. Test Procedure and Results

4.1 Power Rails (Base Power – No Load Switches Enabled)

Test ID	Test Description	Test Point	Expected Value	Measured Value	Pass/Fail	Comments
PWR-01	+12 V Input Verification	TP126	+12.0 V $\pm$ 0.1 V	12.01	Pass	Main board supply
PWR-02	+5 V DC/DC Output (RS6-1205S)	TP123/ TP100	+5.0 V $\pm$ 0.1 V	5.037	Pass	Derived from +12 V supply
PWR-03	3.3V LDO (+3V3)	TP101	+3.3 V $\pm$ 0.05 V	3.311	Pass	IC101 output
PWR-04A	3.3V Driver Rail (for 1V8 Generation)	TP107	+3.3 V $\pm$ 0.05 V	3.314	Pass	IC102 output
PWR-04B	1.8 V LDO (VDD1V8)	TP109	+1.80 V $\pm$ 0.03 V	1.810	Pass	IC103 output

PWR-05A	Pump Supply Verification (input to LT8361)	TP121	+12.0 V $\pm$ 0.1 V	12.01	Pass	
PWR-05B	Pump enable test (R106 = 0 Ω) TP121 = +12 V	TP117	–55 V to –62 V	-59.39	Pass	Negative –60 V Pump LT8361 startup
PWR-05C	Output distribution nodes	TP118/ TP119/ TP122	–55 V to –62 V	-59.39	Pass	

4.2 Power Sequencing – Load Switch Tests

For each load switch, drive GPIO net to 3.3 V to disable rail; leave it floating/0 V to enable rail

Test ID	EN Rail	GPIO Pin	Rail TP	Rail	Expected	Measured	Pass/Fail	LED
SEQ-01	VDD3V3_IO	A3	TP201	VDD3V3	3.3 V	3.311	Pass	LED203
SEQ-02	VDRV3V3_IO	A4	TP203	VDRV3V3	3.3 V	3.311	Pass	LED204
SEQ-03	VDD1V8_IO	D9	TP205	VDD1V8	1.8 V	1.810	Pass	LED206
SEQ-04	VDRV1V8_IO	A2	TP207	VDRV1V8	1.8 V	1.810	Pass	LED207
SEQ-05	VDDA1V8_IO	D6	TP209	VDDA1V8	1.8 V	1.810	Pass	LED208
SEQ-06	VDDCB1V8_IO	D5	TP211	VDDCB1V8	1.8 V	1.810	Pass	LED209

4.3 MCU (ESP32-S3) Bring-up

Test ID	Description	Expected	Pass/Fail	Comments
MCU-01	Power Feather via USB	Board enumerates	Pass	
MCU-02	Power board-side 3V3	Stable +3.3 V at ESP pins	Pass	
MCU-03	Flash test firmware	Serial output OK	Pass	
MCU-04	No brownout resets	Stable operation	Pass	

4.4 Logic Level Shifters (SN74AXC4T774)

Test ID	TP(s)	Signal	Expected	Pass/Fail	Comments
LS-01	TP300/TP302	+3V3_MCU	+3.3V $\pm$ 0.1 V	Pass	
LS-02	TP301/TP303	+1V8	+1.8V $\pm$ 0.1 V	Pass	
LS-03	R310/R314/R319	SCK / MOSI / SS	3.3 V logic input from MCU toggles 3.3 V	Pass	
LS-04	R311/R315/R320	CLKIN / DIN / STDIN	Proper 3.3V -> 1.8 V translation	Pass	

4.6 ASIC Power-up Sequence (Automatic via MCU)

Step	Expected Behavior
1	All Rails are enabled on startup, then all 6 MCU GPIOs are enabled, disabling all +3.3V and +1.8V power rails as well as -60V rail
2	VDD3V3_IO & VDRV3V3_IO enabled
3	VDD1V8_IO, VDRV1V8_IO, VDDA1V8_IO & VDDCB1V8_IO enabled
4	VDD3V3_IO Disabled
5	Wait 2 seconds
6	VDD3V3_IO enabled
7	All status LEDs ON according to rails

5. Oscilloscope Tests

Test ID	Description	TP	Expected		Measured Value		Pass/Fail	Comments
			20M BW	Full BW	20M BW	Full BW		
OSC-01	+5 V ripple	TP100	< 50–80 mVpp		26.97mVpp	57.91 mVpp	Pass	Image 28
OSC-02	+3.3 V ripple	TP102	< 50 mVpp		5.60 mVpp	27.04 mVpp	Pass	Image 30 /Image 29
OSC-03	+1.8 V ripple	TP110	< 30–50 mVpp		5.48 mVpp	30.47 mVpp	Pass	Image 31 /Image 32
OSC-04	–60 V ripple	TP118	< 150–200 mVpp		20.40 mVpp	64.80 mVpp	Pass	Image 33 /Image 34 DC
OSC-05	–60 V ripple	TP118	< 150–200 mVpp		18.40 mVpp	64.82 mVpp	Pass	Image 35 /Image 36 AC

6. Detailed Bring-up Steps

6.1 Visual Inspection

- ☒ Check solder joints, critical IC alignment, DDR3 connector alignment
- ☒ Verify high voltage components match orientation
- ☒ Confirm test points visible & reachable

6.2 Base Power-up

- ☒ Apply +12 V with **current limit 200 mA**
- ☒ Verify +5 V and all LDO outputs
- ☒ Monitor thermal behavior of regulators

6.3 Load Switch Verification

- ☒ Drive each EN pin
- ☒ Measure corresponding rail and LED status
- ☒ Ensure no rail activates unintentionally

6.4 HV Pump Verification

- ☒ Enable pump
- ☒ Confirm 60 V stability

- ☒ ~~Disable pump after test~~

6.5 MCU + SPI Bring-up

- ☒ ~~Flash firmware~~
- ☒ ~~Toggle SPI pins~~
- ☒ ~~Confirm translation on ASIC-side TPs~~

6.6 ASIC Bring-up

- ☒ ~~Execute power sequence~~
- ☒ ~~Write known registers~~
- ☒ ~~Verify pixel enable~~
- ☐ Measure SMA outputs

6.7 Final Verification

- ☒ ~~Cycle power 10*~~
- ☒ ~~Confirm stable operation & sequencing~~
- ☒ ~~Check all rails and SPI on cold/warm boot~~